

Cyanobacterial Blooms and the Occurrence of the neurotoxin beta-N-methylamino-L-alanine (BMAA) in South Florida Aquatic Food Webs

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Recent studies demonstrate that most cyanobacteria produce the neurotoxin beta-N-methylamino-L-alanine (BMAA) and that it can biomagnify in at least one terrestrial food chain. BMAA has been implicated as a significant environmental risk in the development of neurodegenerative diseases such as Alzheimer's disease, Parkinson's disease, and Amyotrophic Lateral Sclerosis (ALS). We examined several blooms of cyanobacteria in South Florida, and the BMAA content of resident animals, including species used as human food. A wide range of BMAA concentrations were found, ranging from below assay detection limits to approximately 7000 µg/g, a concentration associated with a potential long-term human health hazard.